Machine Learning

Naïve Bayes & Bayesian Network

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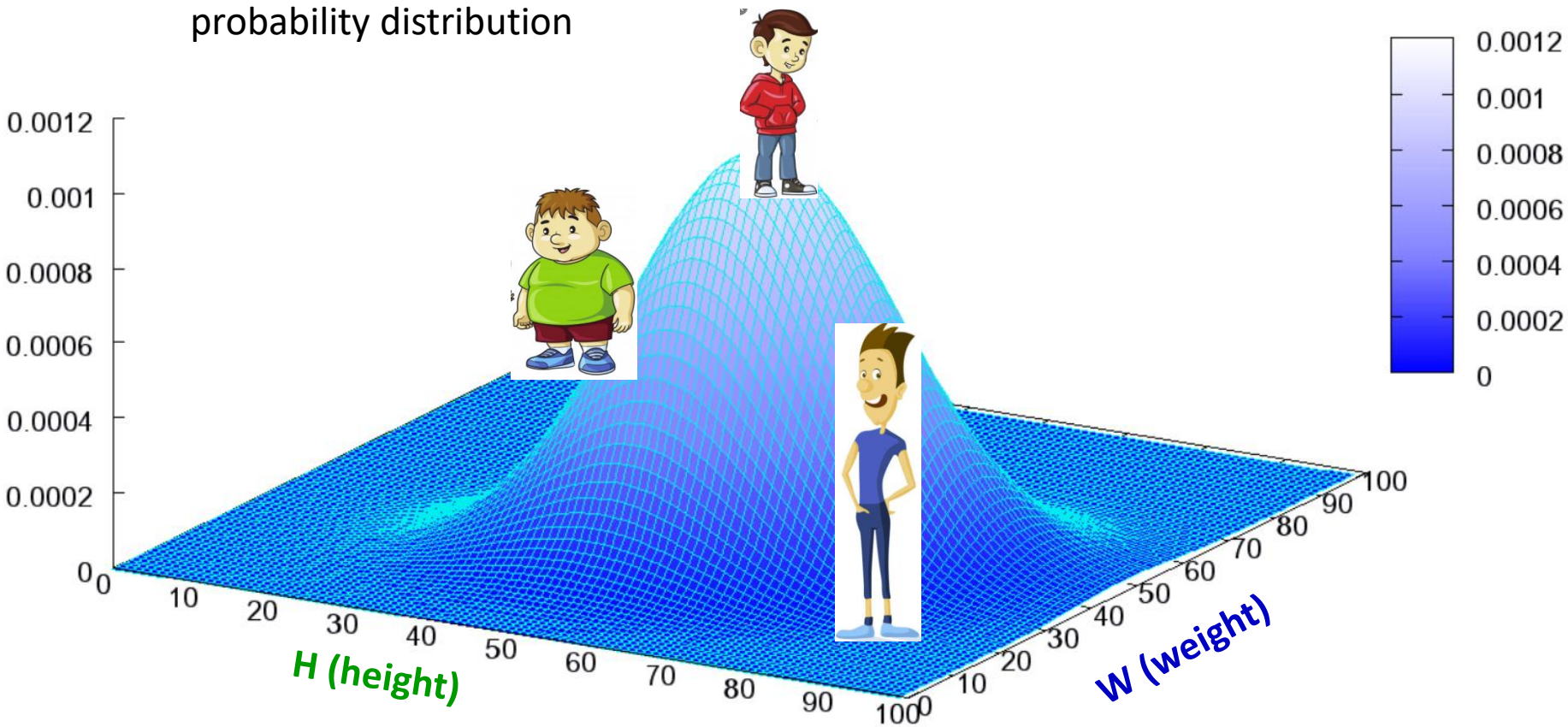
Contents

- **Background: Probability basics**
  - Joint probability
  - Bayes rule
  - Conditional independence (or, naïve bayes assumption)
- **Bayesian decision theory**
- **Naïve Bayes classifier**
- **Bayesian networks**

Joint Probability Distribution

What is joint probability distribution?

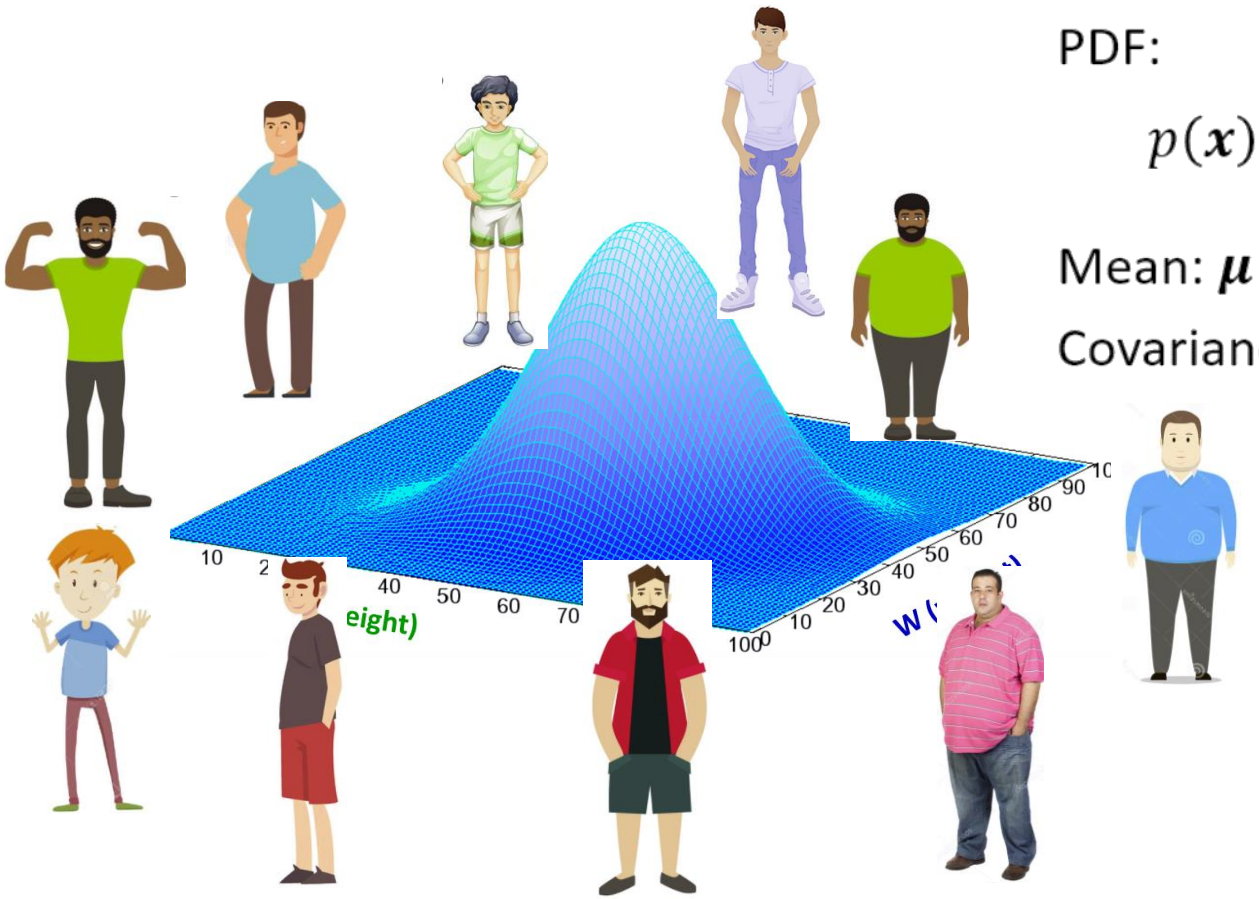
- A probability distribution, P(X), is the function that maps the input X to the probabilities
  - If X is a single variable => P(X) is a probability distribution
  - If X is a set of multiple variables, e.g., X={**H (height), W (weight)**} => P(X) is a joint probability distribution



Joint Probability Distribution

Joint probability for the continuous domain

- Fit your chosen probability density function (e.g., Gaussian distribution) by using your training data



PDF:

$$p(\boldsymbol{x}) = \frac{1}{\left|\sqrt{2\pi\boldsymbol{\Sigma}}\right|} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

Mean: $\boldsymbol{\mu}$ (vector 2x1)

Covariance: $\boldsymbol{\Sigma}$ (matrix 2x2)

Joint Probability Distribution

Joint probability for the discrete domain

- Fill out the **joint probability table** by enumerating the number of examples from your training data
  - S=season
  - T=temp
  - W=weather

S	T	W	P
Summer	Hot	sun	0.30
Summer	Hot	Rain	0.05
Summer	Cold	Sun	0.10
Summer	Hot	Rain	0.05
Winter	Hot	Sun	0.10
Winter	Hot	Rain	0.05
Winter	Cold	sun	0.15
Winter	Cold	Rain	0.20

Bayes' Theorem

- **A joint distribution $\boldsymbol{P(X,Y)}$ can be factorized as:**

$$P(X,Y) = P(Y \mid X)P(X) = P(X \mid Y)P(Y)$$

- **Rearrangement:**

likelihood

posterior $\leftarrow$ $$P(Y \mid X) = \frac{P(X|Y)}{P(X)} P(Y)$$ $\rightarrow$ prior

evidence

- **Usage:**
  - Bayes' rule allows us to build one conditional from its inverse
  - Useful when $P(Y|X)$ is harder to get than $P(X|Y)$

Joint Probability Table

- **Estimated joint probability** $P(X, Y)$
  - It allows to infer any desired probability
    => e.g., **conditional** from joint: $P(Y|X) = \dfrac{P(X,Y)}{P(X)}$

S	T	W	P
Summer	Hot	sun	0.30
Summer	Hot	Rain	0.05
Summer	Cold	Sun	0.10
Summer	Hot	Rain	0.05
Winter	Hot	Sun	0.10
Winter	Hot	Rain	0.05
Winter	Cold	sun	0.15
Winter	Cold	Rain	0.20

Joint probability table $\boldsymbol{P(S,T,W)}$

- $P(W)$

$$P(\text{sun}) = 0.30 + 0.10 + 0.10 + 0.15 = 0.65$$
$$P(\text{rain}) = 0.05 + 0.05 + 0.05 + 0.20 = 0.35$$

- $P(W|\textbf{winter})$

$$P(\text{sun}|\text{winter}) = \frac{0.10 + 0.15}{0.5} = 0.5$$

$$P(\text{rain}|\text{winter}) = \frac{0.05 + 0.20}{0.5} = 0.5$$

- $P(W|\textbf{winter, hot})$

$$P(\text{sun}|\text{winter, hot}) = \frac{0.10}{0.15} = \frac{2}{3}$$
$$P(\text{rain}|\text{winter, hot}) = \frac{0.05}{0.15} = \frac{1}{3}$$

Inference from Joint: Generalization

- **Given n variables:** $X_1, X_2, \dots X_n$
- **We want**

$$P(Y|e_1, \dots, e_k)$$

  - Observed evidences: $e_1, \dots, e_k$
  - Target variable: Y
  - Hidden variables: $H_1, \dots, H_r$

- **Example**

	$P(W)$	$P(W\|\text{winter})$	$P(W\|\text{winter}, \text{hot})$
Evidence	None	$S = \text{winter}$	$S = \text{winter}$, $T = \text{hot}$
Target	W	W	W
Hidden	S, T	T	None

Inference from Joint: Generalization

Steps for inference of conditional probability:

(1) $P(X, H, Y) \Rightarrow P(X, Y)$, (2) $P(Y|X) = \dfrac{P(X, Y)}{P(X)}$

- **(1) Sum out hidden H** to get **joint** of target and evidence => **marginalization!**

$$P(Y, e_1, \dots, e_k) = \sum_{h_1 \dots, h_r} P(Q, h_1, \dots h_r, e_1, \dots, e_k)$$

- (2) **Normalization** => computing normalization factor Z:

$$Z = \sum_{y} P(Y, e_1, \dots, e_k)$$

- Then,

$$P(Y|e_1, \dots, e_k) = \frac{1}{Z} P(Y, e_1, \dots, e_k)$$

Inference from Joint: Generalization

- **Example: $P(W \mid$ "winter" $)$**
  - Evidence: $S = \text{winter}$
  - Target: W
  - Hidden T

S	T	W	P
Summer	Hot	sun	0.30
Summer	Hot	Rain	0.05
Summer	Cold	Sun	0.10
Summer	Hot	Rain	0.05
Winter	Hot	Sun	0.10
Winter	Hot	Rain	0.05
Winter	Cold	sun	0.15
Winter	Cold	Rain	0.20

- **Step 1: select** $\Rightarrow P(S = \text{winter}, T, W)$

S	T	W	P
Winter	Hot	Sun	0.10
Winter	Hot	Rain	0.05
Winter	Cold	Sun	0.15
Winter	Cold	Rain	0.20

- **Step 2: marginalize T** $\Rightarrow P(\text{winter}, W)$

S	T	W	P
Winter	Hot/Cold	Sun	0.25
Winter	Hot/Cold	Rain	0.25

- **Step 3: normalize** $\Rightarrow P(W \mid \text{winter})$

S	T	W	P
Winter	Hot/Cold	Sun	0.5
Winter	Hot/Cold	Rain	0.5

Complexity

Is it feasible?

- Assume: n variables with domain size d
- Worst-case time complexity: $O(d^n)$
- Space complexity to store the joint distribution: $O(d^n)$

=> Too expensive

How to reduce the size of joint probability table?

- Independence $\Rightarrow$ Too much strong assumption
- **Conditional Independence** $\Rightarrow$ **Naïve Bayes, Bayes net**
- Markov Property $\Rightarrow$ Hidden Markov Model (HMM)
- Latent Variable $\Rightarrow$ Mixture Model, Variational AE

Two Families of Conditional Independence

- **Naïve Bayes Classifier**
  - Conditional independence
  - Bayesian decision theory
  - Naïve Bayes classification

- **Bayesian Networks (Bayes Net)**
  - Graphical model
  - Probability representation by Bayes net
  - D-separation

Independence

- **Two variables are *independent* if**

$$\forall x, y : P(x, y) = P(x)P(y)$$

i.e. Their joint distribution *factors* into a product of two simpler distributions

  - We write $X \perp\!\!\!\perp Y$

- **Independence: it simplifies our probabilistic task**

- **Example: assume n Bernoulli random variables $\boldsymbol{X_1, X_2, \ldots, X_n}$**
  - # Entries in their joint table $P(X_1, X_2, \ldots, X_n)$?
    - In general (no independence assumption) $\Rightarrow$ $\boxed{2^n}$
    - All independent $\Rightarrow$ $2n$

- **Limitation: *too strong* in practice**

Conditional Independence

- **Alternative: *conditional independence* (CI)**
  - Still strong but more readily applicable to real world problems

- **Most basic & robust form of knowledge on uncertain environments**
- **X is conditionally independent of Y given C** $\boxed{X \perp\!\!\!\perp Y \mid C}$
  - If and only if

$$\forall x, y, c : P(x, y \mid c) = P(x \mid c)P(y \mid c)$$

or, equivalently,

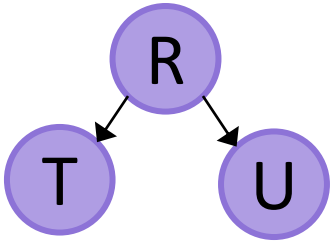
$$\forall x, y, c : P(x \mid c, y) = P(x \mid c)$$

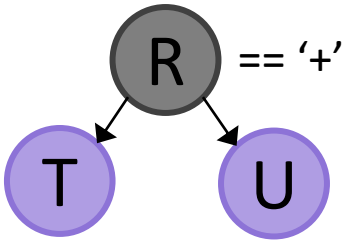
- **How CI helps?**
  - Enables **compact representation** of joint probability tables

Example: Traffic, Umbrella & Raining

- **Variables**
  - T: traffic (+, -)
  - U: umbrella (+, -)
  - R: rain (+, -)

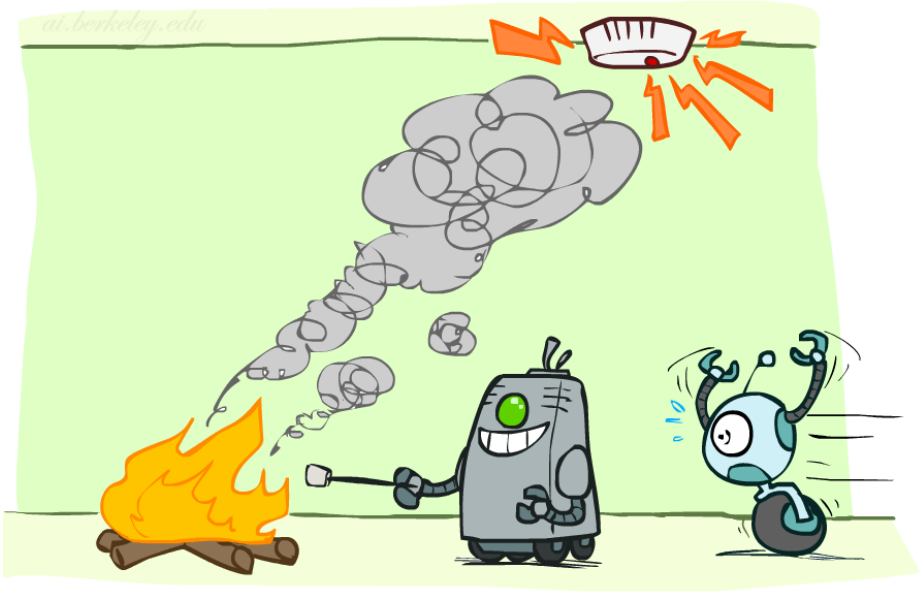
- **One possible CI assumption:** $T \perp\!\!\!\perp U \mid R$
  - If R is unobserved, then F and A are not independent
  - If R='+' (**observed**), then T and U are **conditionally independent**



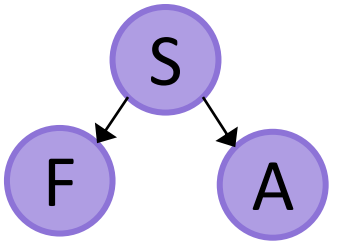


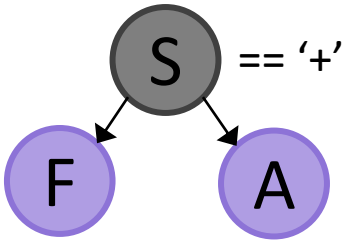
Example: Fire, Smoke & Alarm

- **Variables**
  - F: fire (+, -)
  - S: smoke (+, -)
  - A: alarm (+, -)



- **One possible CI assumption: $F \perp\!\!\!\perp A \mid S$**
  - If S is unobserved, then F and A are not independent
  - If S='+' (**observed**), then F and A are **conditionally independent**





Conditional Independence Assumption

- **Also known as *naïve Bayes* assumption: given class C_k,**
  - Feature x_i is conditionally independent of every other feature x_j for $j \neq i$

$$P(x_i, x_j \mid C_k) = P(x_i \mid C_k)\, P(x_j \mid C_k)$$

- **Meaning:**
  - If class label is given, knowledge of x_j has no effect on our beliefs about value of x_i

$\Rightarrow$ **still strong** assumption, **but works well** in many problems

Bayesian Decision Theory

Recall Bayes rule

likelihood

posterior $\leftarrow$ $$P(C \mid X) = \frac{P(X|C)}{P(X)} P(C)$$ $\rightarrow$ prior

evidence

General usage: probabilistic classification

- $p(C_k|x)$ (posterior): Conditional probability of class $k \in K$ given input X
- The most probable is chosen as the classifier $h(X)$:

$$h(X) = \arg\max p(C_k|X)$$

Example: Spam Mail Filtering

Data

	Label	SMS
0	spam	SECRET PRIZE! CLAIM SECRET PRIZE NOW!!
1	ham	Coming to my secret party?
2	spam	Winner! Claim secret prize now!

Variables

- Each email is represented by feature vector (here, feature = word)

$$P(w_1, w_2, w_3, \dots, w_n | \, C_{spam})$$

e.g. common words for spam email: drug, buy, lottery, win, …

	Label	secret	prize	claim	now	coming	to	my	party	winner
0	spam	2	2	1	1	0	0	0	0	0
1	ham	1	0	0	0	1	1	1	1	0
2	spam	1	1	1	1	0	0	0	0	1

Target variable C:

- $C = C_1$ for spam
- $C = C_2$for normal (ham)
- $P(C_1)$: a prior that the email is spam
- $P(C_2)$: a prior that the email is normal

Example: Spam Mail Filtering

- **Prior: it reflects our knowledge of how likely each type of email will appear**
  - Before we actually see it

- **How to choose priors $P(C_1)$ and $P(C_2)$?**
  - Equal priors (0.5), or may use different values depending on service provider

- $\boldsymbol{P(C_1) + P(C_2) = 1}$

- **How about classifying only based on priors?**
  - Decision rule: taking actions based on observation on target variable C

$$\text{decide} \begin{cases} C_1, & \text{if } P(C_1) > P(C_2); \\ C_2, & \text{otherwise} \end{cases}$$

Example: Spam Mail Filtering

- **Let's use input variable w**

- **We can use Bayes' Theorem to express posterior with prior $P(C_j)$ and likelihood $p(\mathbf{w}|C_j)$:**

$$P(C_j|\mathbf{w}) = \frac{p(\mathbf{w}|C_j)P(C_j)}{p(\mathbf{w})}$$

- **Given an email w=[$w_1, w_2, \ldots, w_n$]^T**
  - its class is predicted as:

$$\begin{aligned} C_k &= \underset{C_j}{\operatorname{argmax}}\, P(C_j|\,\mathbf{w}) \\ &= \underset{C_j}{\operatorname{argmax}} \frac{P(\mathbf{w}|\,C_j)P(C_j)}{P(\mathbf{w})} \\ &= \underset{C_j}{\operatorname{argmax}}\; P(\mathbf{w}\mid C_j)P(C_j) \\ &= \underset{C_j}{\operatorname{argmax}}\, P(w_1, w_2, \ldots, w_n|C_j)\, P(C_j) \end{aligned}$$

Bayesian Classification Summary

- **Decision making:**

$$C^* = \arg\max_j P\left(C_j \middle| \mathbf{w}\right)$$

That is, decide $\begin{cases} C_1, & \text{if } P(C_1|\mathbf{w}) > P(C_2|\mathbf{w}) \\ C_2, & \text{otherwise} \end{cases}$

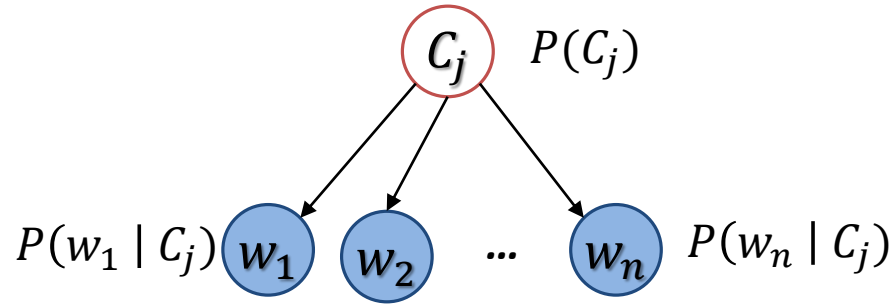
- **Decision with likelihood and prior:**

$$decide \begin{cases} C_1, & \text{if } P(\mathbf{w}|C_1)P(C_1) > P(\mathbf{w}|C_2)P(C_2) \\ C_2, & \text{otherwise} \end{cases}$$

Naïve Bayes (NB) Classifier

- **Under the *NB assumption*,**

$$P(w_1, w_2, \ldots, w_n | C_j) = \prod_{i=1}^{n} P(w_i \mid C_j)$$



- **Naïve Bayes classifier predicts an email w = $[w_1, w_2, \ldots, w_n]^T$ by:**

$$C_k = \underset{C_j}{\operatorname{argmax}} \prod_{i=1}^{n} P(w_i \mid C_j)\, P(C_j)$$

- **Estimating each $P(w_i \mid C_j)$ instead of $P(w_1, w_2, \ldots, w_n | C_j)$**
  - Greatly **reduces the number of computations**

Properties of Naïve Bayes Classifier

- **Under the *NB assumption*, drug ⫫ buy | spam**

$$P(\text{drug}, \text{buy} \mid C_{spam}) = P(\text{drug} \mid C_{spam}) P(\text{buy} \mid C_{spam})$$

  - Once we know it is a spam, knowledge of whether "buy" appears in the email has **no effect** on our belief about whether "drug" appears
  - Computing joint probability table for $P(w_1, w_2, w_3, \ldots, w_n \mid C_{spam})$ is difficult, but computing each $P(w_1 \mid C_{spam})$, $P(w_2 \mid C_{spam})$, …, $P(w_n \mid C_{spam})$ is easy.

- **Fast estimation**
  - Estimating $P(x_i \mid C_j)$ and $P(C_j)$ based on frequencies in training data

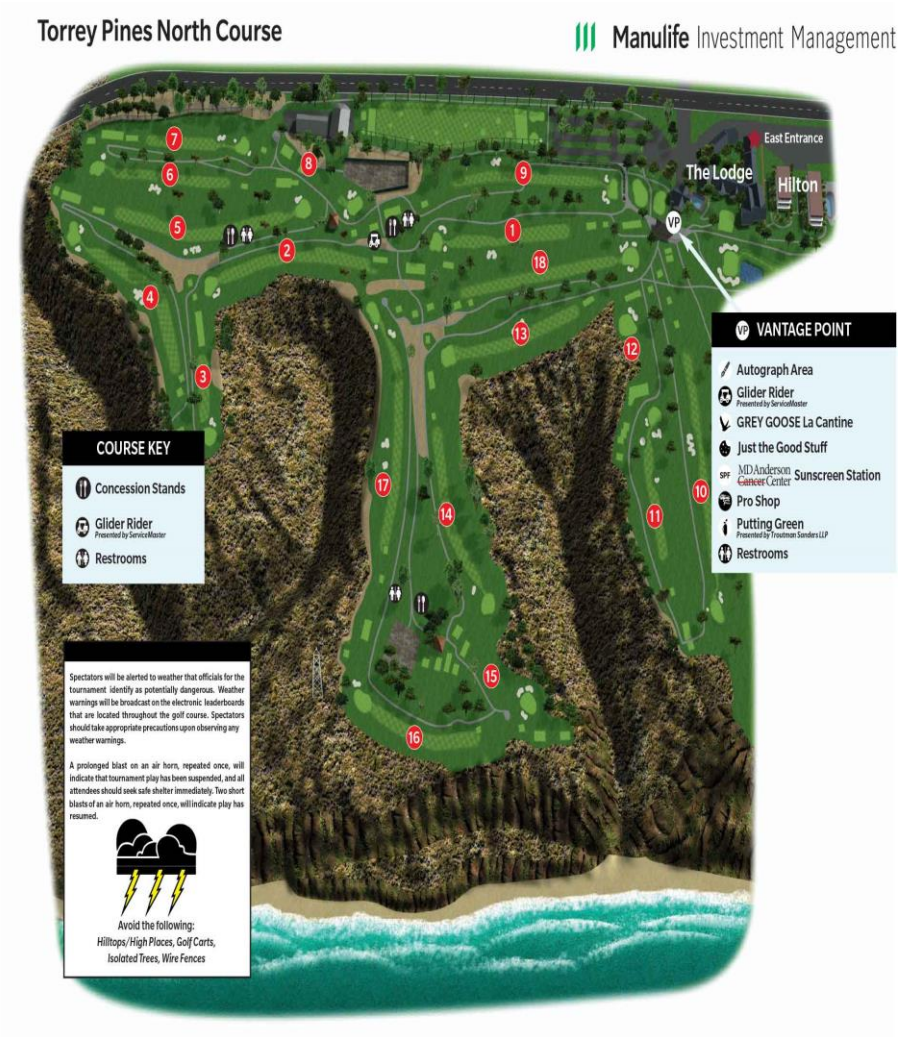
- **Despite strong assumption of conditional independence,**
  - NB tends to work well in practice

  ⇒ Quite competitive with other classification methods

Example: Weather Conditions for Playing Golf

☐ Play record (aka training data)

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no



Example: Weather Conditions for Playing Golf

- Given the following weather condition, play or not?

Outlook	Temperature	Humidity	Windy
sunny	cool	high	true

- Make a probability table

Outlook			Temperature			Humidity			Windy		
	yes	no		yes	no		yes	no		yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2
overcast	4	0	mild	4	2	normal	6	1	true	3	3
rainy	3	2	cold	3	1						

Example: Weather Conditions for Playing Golf

- **Compute priors**
  - # Play_yes = 9, #Play_no = 5
  - P (Play = yes) = $\frac{9}{14}$ $\quad$ P (Play = no) = $\frac{5}{14}$
- **Compute each likelihood:**

Outlook			Temperature			Humidity			Windy		
	yes	no		yes	no		yes	no		yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2
overcast	4	0	mild	4	2	normal	6	1	true	3	3
rainy	3	2	cold	3	1						

P (Windy = true | Play = yes) = $\frac{3}{9}$

P (Windy = true | Play = no) = $\frac{3}{5}$

....

Example: Weather Conditions for Playing Golf

Query data

Outlook	Temperature	Humidity	Windy
sunny	cool	high	true

Multiply each likelihood:

$P(O = \text{sunny} \mid \textbf{yes})\ P(T = \text{cool} \mid \textbf{yes})\ P(H = \text{high} \mid \textbf{yes})\ P(W = \text{true} \mid \textbf{yes})$

$$= \frac{2}{9} \cdot \frac{3}{9} \cdot \frac{3}{9} \cdot \frac{3}{9} = \frac{6}{729} = 0.00823$$

$P(O = \text{sunny} \mid \textbf{no})\ P(T = \text{cool} \mid \textbf{no})\ P(H = \text{high} \mid \textbf{no})\ P(W = \text{true} \mid \textbf{no})$

$$= \frac{3}{5} \cdot \frac{1}{5} \cdot \frac{4}{5} \cdot \frac{3}{5} = \frac{36}{625} = 0.0576$$

Example: Weather Conditions for Playing Golf

- **Decision:**

$$\text{class} \quad \Rightarrow \quad \underset{\text{Play} \in \{\text{yes, no}\}}{\operatorname{argmax}} \left\{ P(\text{yes}) \cdot \frac{6}{729}, \quad P(\text{no}) \cdot \frac{36}{625} \right\}$$

$$= \underset{\text{Play} \in \{\text{yes, no}\}}{\operatorname{argmax}} \left\{ \frac{9}{14} \cdot \frac{6}{729}, \quad \frac{5}{14} \cdot \frac{36}{625} \right\}$$

$$= \underset{\text{Play} \in \{\text{yes, no}\}}{\operatorname{argmax}} \{0.0053, \ \mathbf{0.0206}\}$$

decision: <u>No!</u>

Bayesian Networks

Main idea:

- Describe complex joint distributions (models) by using simple, **local** distributions (conditional probabilities)
- Describe how variables **locally** interact via graph structures
- Local interactions **chain** together $\Rightarrow$ give **global**, indirect interactions

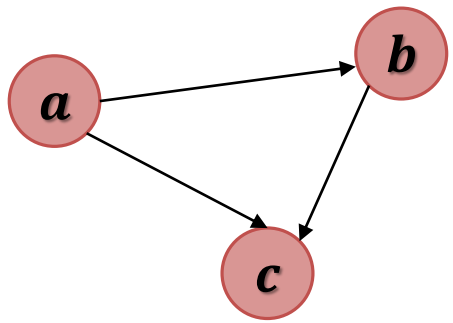
Bayesian Network VS. Naïve Bayes Classifier

- Both are exploiting the conditional independence among variables
- NB classifier is tailored to do classification
- Bayes net is more interested in capturing (joint) data distribution

Bayesian Networks

- **Problems with using full joint distribution tables**
  - The joint distribution table is often **too big** to represent explicitly
  - Bayes net can be a better way of representing such joint distribution

- **Graphical model: it models relationships among variables**

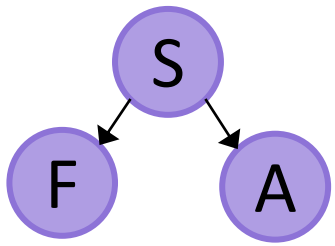


$$P(a,b,c) = P(a)P(b|a)P(c|a,b)$$

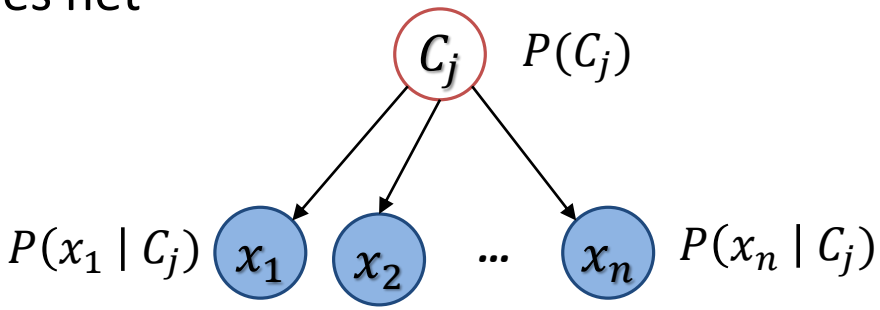
Bayesian Networks

- **Bayesian networks is a probabilistic, directed graphical model**
  - Can express relationships between variables

- **Conditional independence: Fire $\perp\!\!\!\perp$ Alarm | Smoke**



- **Naïve Bayes model:** $P(x_1, x_2, \ldots, x_d | C_j) = \prod_{i=1}^{n} P(x_i \mid C_j)$
  - A special case of Bayes net

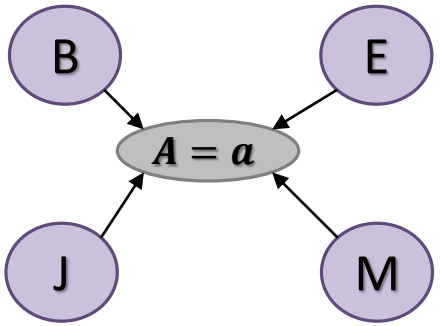


Architecture

- **Node = a random variable**
  - Can be assigned (observed) or unassigned (unobserved)

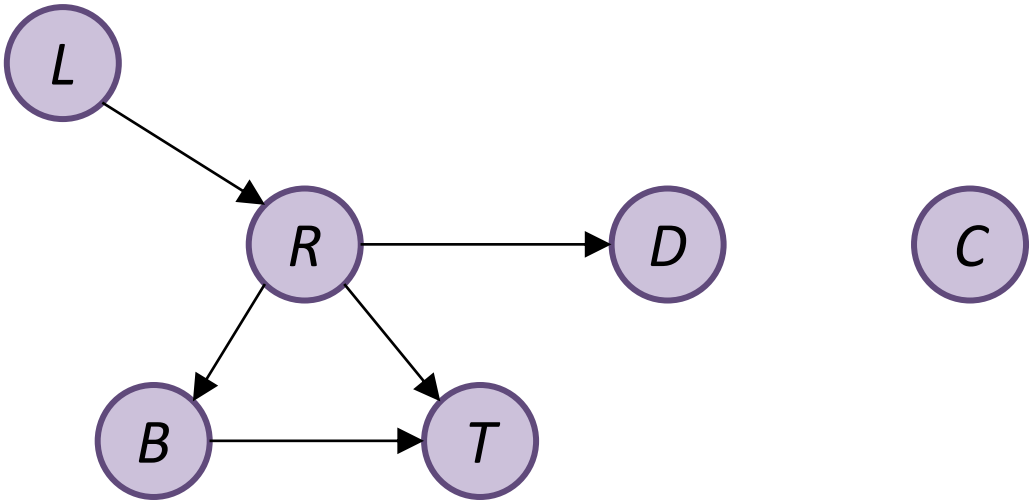
- **Edge = probabilistic relationships between random variables**
  - Indicates direct influence between variables

- **A directed acyclic graph (DAG)**
  - No directed cycles
  - Comprises nodes connected by edges



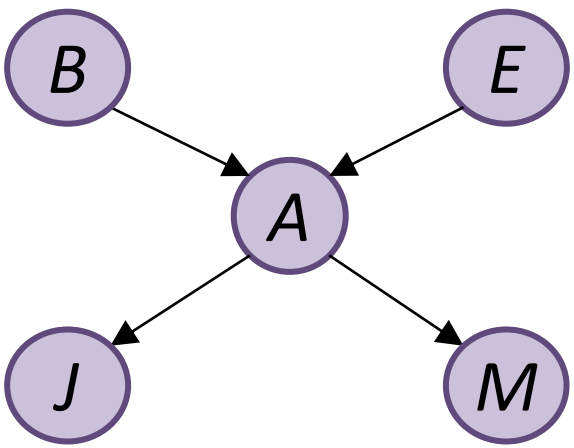
Example

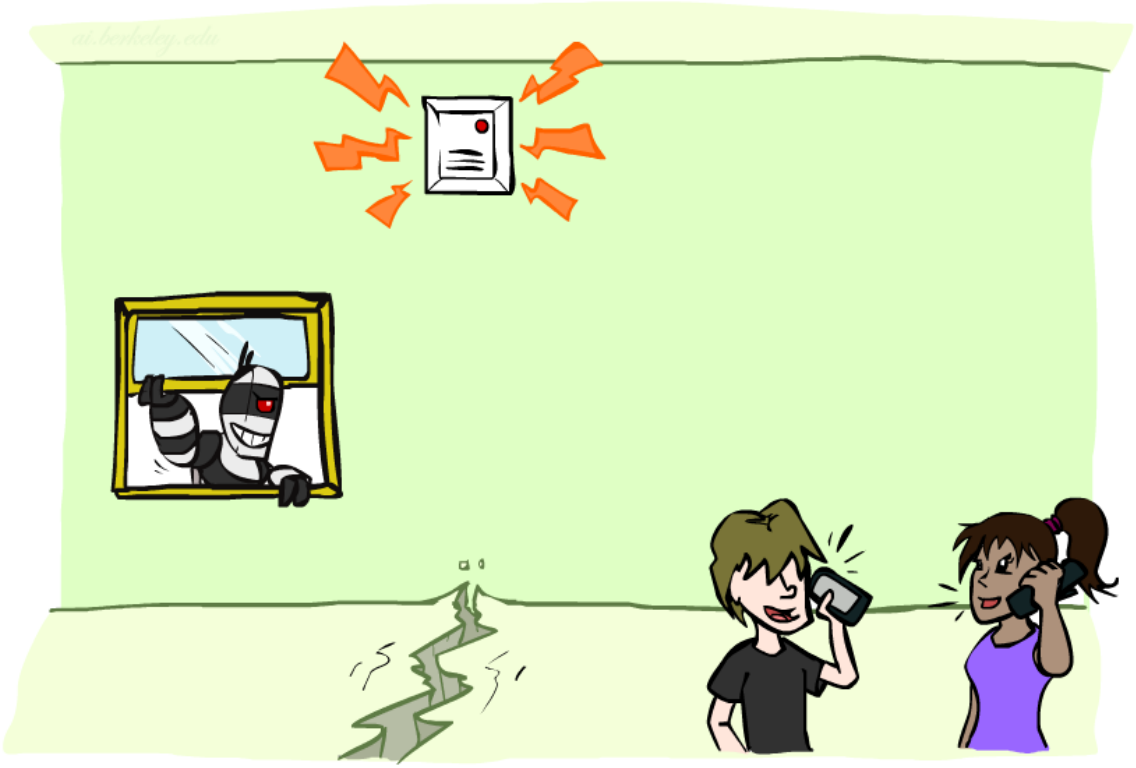
- **Example: traffic**
  - T: traffic
  - R: rain
  - L: low pressure
  - D: roof drips
  - B: ballgame
  - C: cavity



Example

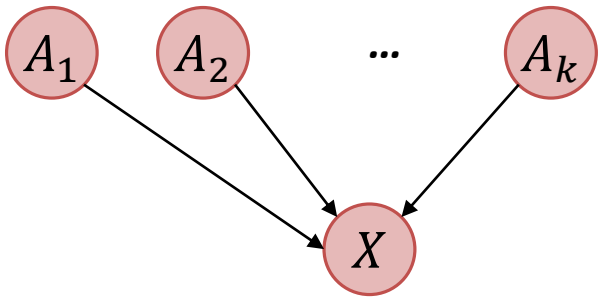
- **Example: alarm network**
  - B: burglary
  - A: alarm goes off
  - M: Mary calls
  - J: John calls
  - E: earthquake





Local Conditional Distribution

- **Each node X in the Bayes net is associated with a conditional distribution**



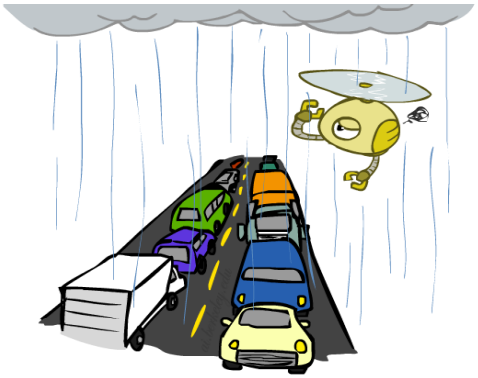
- **Then, the conditional distribution for node X can be written as:**

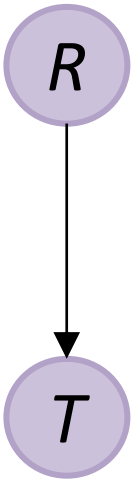
$$P(X \mid \textbf{parents}(X)) = P(X \mid A_1, A_2, \dots, A_k)$$

, where parents$(X) = \{A_1, A_2, \dots, A_k\}$

Local Conditional Distribution

- **Example: traffic**
  - R: rain
  - T: traffic





$\boldsymbol{R}$	$\boldsymbol{P(R)}$
$+r$	0.25
$-r$	0.75

⇒ Node R: encoding prior $P(R)$

$\boldsymbol{R}$	$\boldsymbol{T}$	$\boldsymbol{P(T \mid R)}$
$+r$	$+t$	0.75
$+r$	$-t$	0.25
$-r$	$+t$	0.5
$-r$	$-t$	0.5

⇒ Node T: encoding the conditional probability $P(T|R)$

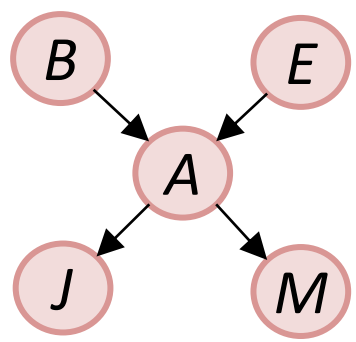
Global Joint Distribution

- **Bayes net implicitly encodes joint distributions**
  - As a **product** of local conditional distributions
- **A Bayes net with N variables $\{X_1, X_2, \dots, X_N\}$**
  - Represents a joint distribution

$$P(\boldsymbol{X}) = P(X_1, X_2, \dots, X_N) = \prod_{i=1}^{N} P(X_i \mid \text{parents}(X_i))$$

  - Multiply all the relevant conditionals together
- **Example**

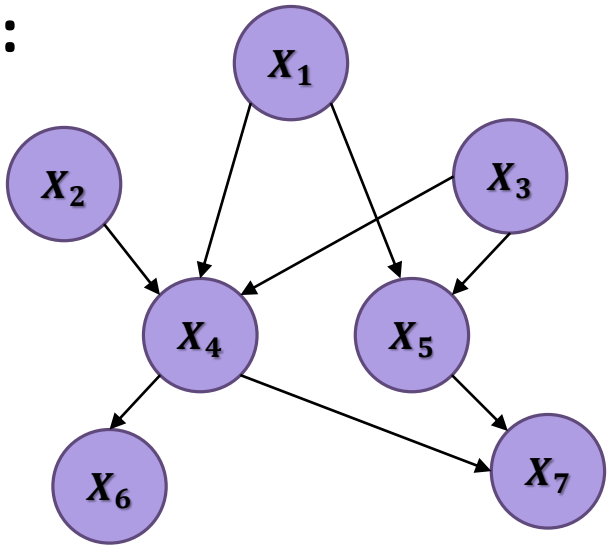
$$P(B, E, A, J, M) = P(B)\, P(E)\, P(A \mid B, E)\, P(J \mid A)\, P(M \mid A)$$



Global Joint Distribution

- **The graph captures how the joint distribution can be decomposed into a product of factors**
  - Where each factor depends only on a subset of the variables

- **Example:**

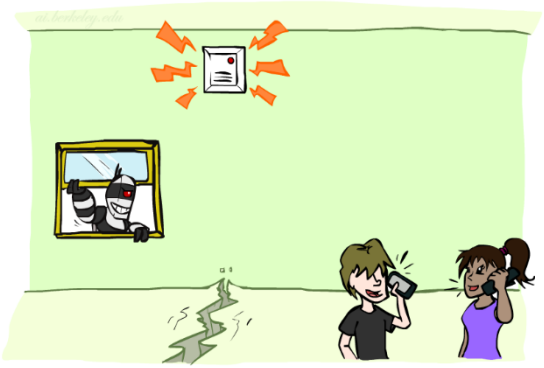


$$P(X_1, X_2, \dots, X_7) =$$
$$P(X_1)\, P(X_2)\, P(X_3)\, P(X_4|X_1, X_2, X_3)\, P(X_5|X_1, X_3)\, P(X_6|X_4)\, P(X_7|X_4, X_5)$$

Example: Alarm Network

- **Question: what is the probability of $P(+b, -e, +a, -j, +m)$?**

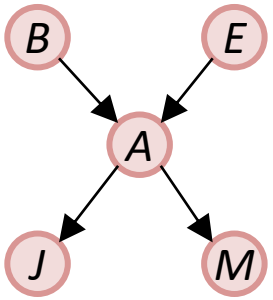
$$P(+b, -e, +a, -j, +m)$$
$$= P(+b)P(-e)P(+a \mid +b, -e)P(-j \mid +a)P(+m \mid +a)$$
$$= 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7$$



B	$P(B)$
$+b$	0.001
$-b$	0.999

E	$P(E)$
+e	0.002
$-e$	0.998

B	E	A	$P(A \mid B, E)$
$+b$	$+e$	$+a$	0.95
$+b$	$+e$	$-a$	0.05
$+b$	$-e$	$+a$	0.94
$+b$	$-e$	$-a$	0.06
$-b$	$+e$	$+a$	0.29
$-b$	$+e$	$-a$	0.71
$-b$	$-e$	$+a$	0.001
$-b$	$-e$	$-a$	0.999



A	J	$P(J \mid A)$
$+a$	$+j$	0.9
$+a$	$-j$	0.1
$-a$	$+j$	0.05
$-a$	$-j$	0.95

A	M	$P(M \mid A)$
$+a$	$+m$	0.7
$+a$	$-m$	0.3
$-a$	+m	0.01
$-a$	−m	0.99

Complexity

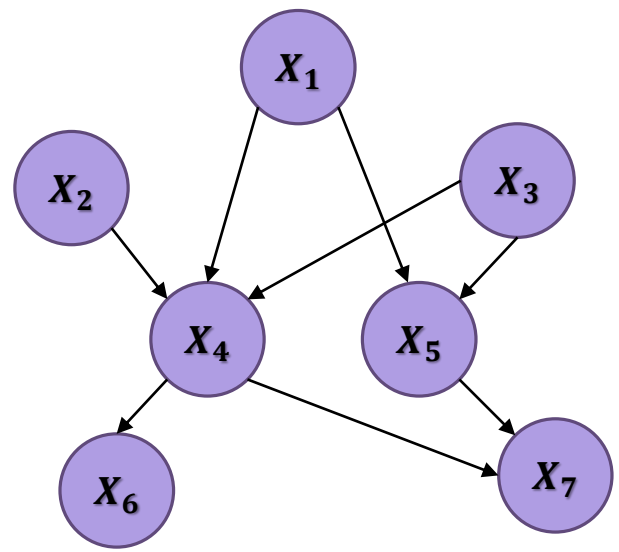
- **Assume the domain size is d, how big is a joint distribution over n variables?**
  - d^n

- **How big is an n-node Bayes net if nodes have up to k parents?**
  - $O(n \times d^{k+1})$: Huge space savings (especially if $k \ll n$)

- **Example:**

$$\boldsymbol{d + d + d + d^4 + d^3 + d^2 + d^3}$$

$$= O(d^4) \ll O(d^7)$$



D-Separation

- **Bayes net topology encodes conditional independence (CI) among variables**
- **Question:**
  - In a given Bayes net, are two variables **X** and **Y** **CI**, given evidence **Z**?
  - If the answer is Yes, we say that $\boldsymbol{X}$ and $\boldsymbol{Y}$ are "**d-separated**" by $\boldsymbol{Z}$

    'd' stands for 'directed'
- **How to determine CI?**
  - Mathematically determine whether $P(X, Y \mid Z) == P(X \mid Z)P(Y \mid Z)$
    - Too many computations required
  - **Alternative: verifying three canonical cases through graphical structures**
    - Causal chain (head-to-tail)
    - Common cause (tail-to-tail)
    - Common effect (head-to-head or v-structure)

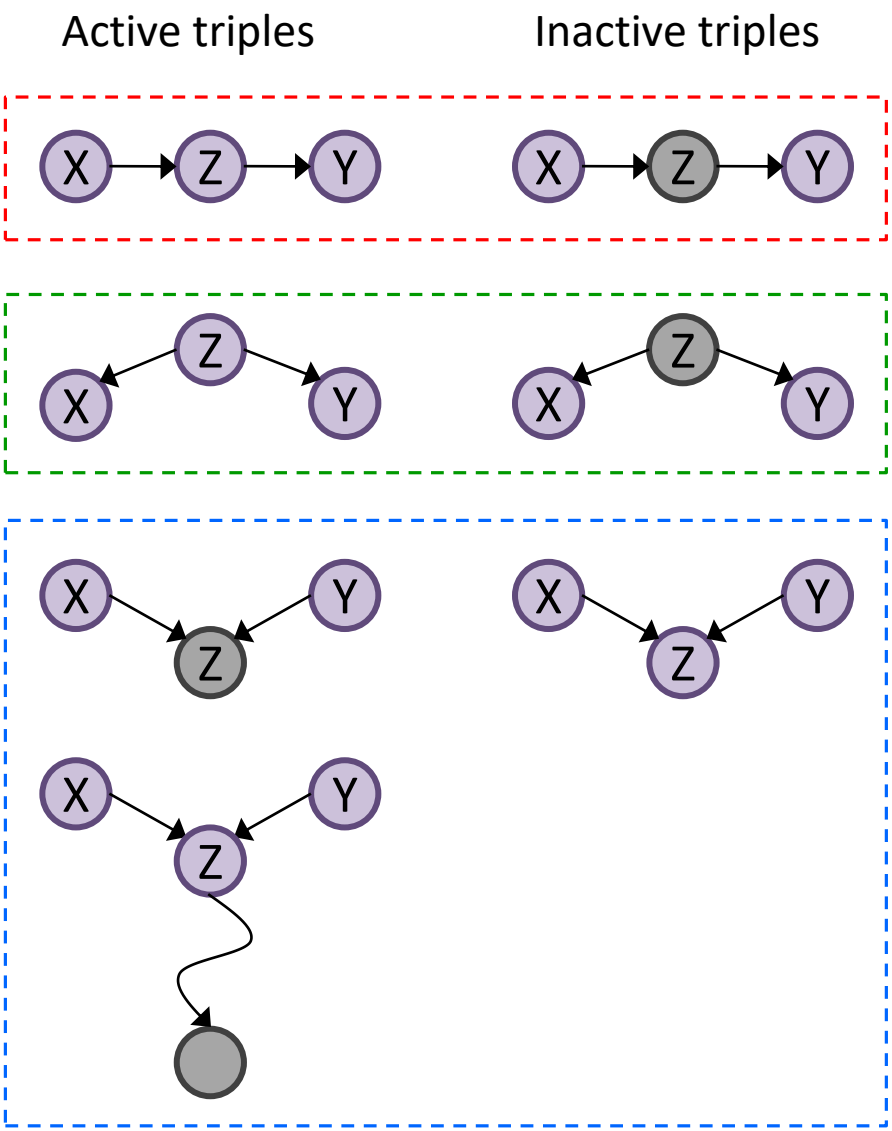
D-Separation

- **Question:**
  - Are $\boldsymbol{X}$ and $\mathbf{Y}$ CI given evidence $\mathbf{Z}$?

- **Consider all (undirected) paths from $\boldsymbol{X}$ to $\boldsymbol{Y}$**

  $\Rightarrow$ **no** active paths $\Leftrightarrow$ independence

- **A path is active if:**
  - Causal chain: $\boldsymbol{X} \rightarrow \boldsymbol{Z} \rightarrow \boldsymbol{Y}$
    - when Z is **unobserved**
  - Common cause: $\boldsymbol{X} \leftarrow \boldsymbol{Z} \rightarrow \boldsymbol{Y}$
    - when Z is **unobserved**
  - Common effect: $\boldsymbol{X} \rightarrow \boldsymbol{Z} \leftarrow \boldsymbol{Y}$
    - when Z or its descendant is **observed**



D-separation

- **More general query: is the following implied by a Bayes net?**

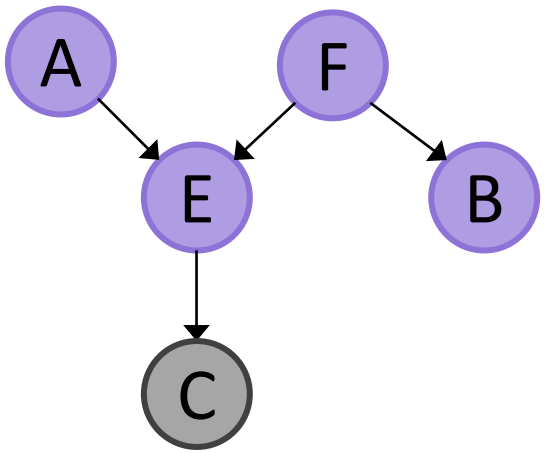
$$\mathcal{A} \perp\!\!\!\perp \mathcal{B} \mid \mathcal{C}$$

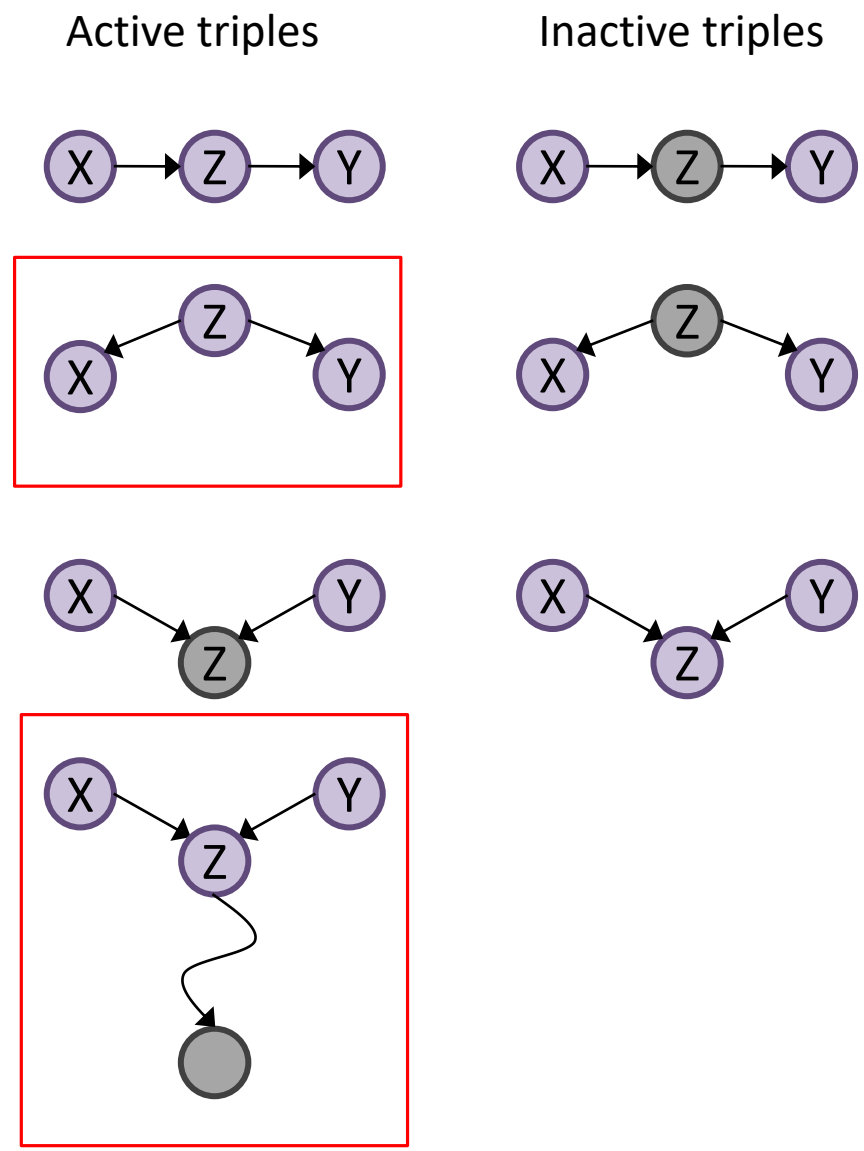
observed

  - $\mathcal{A}, \mathcal{B}, \mathcal{C}$: arbitrary non−intersecting set of variables

- **Check all (undirected) paths from any node $A \in \mathcal{A}$ to any node $B \in \mathcal{B}$**
  - If one or more path is **active**
    - Conditional independence is **not guaranteed**
  - Otherwise: all paths are **inactive**
    - Conditional independence is **guaranteed**
    - Then, we say: $\mathcal{A}$ is **d−separated** from $\mathcal{B}$ by $\mathcal{C}$

Example

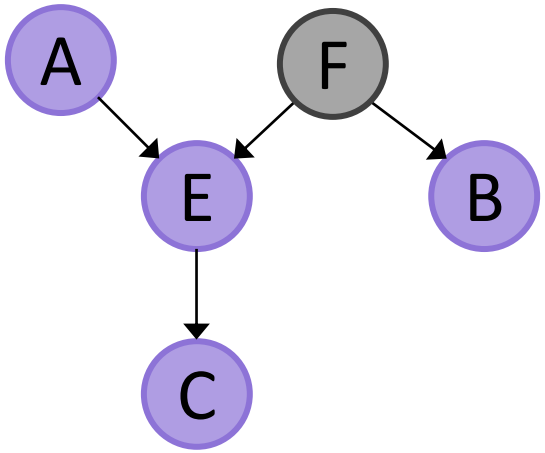


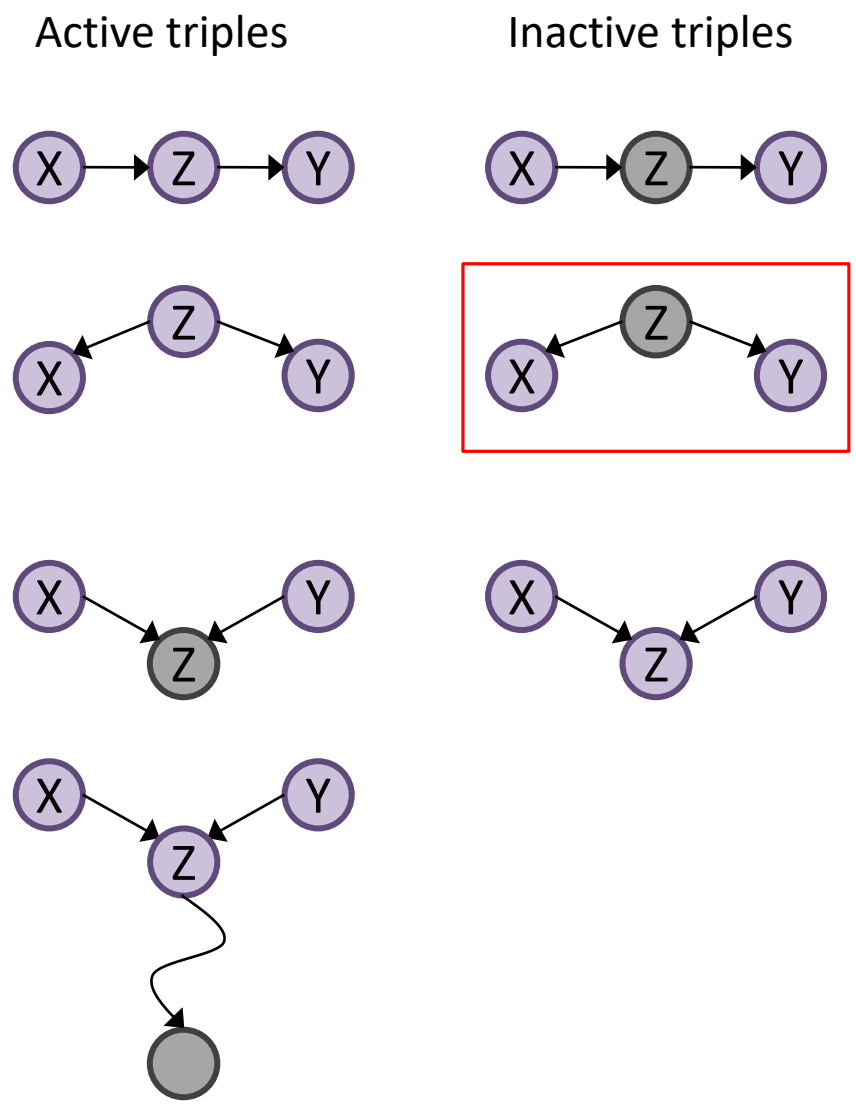


- **Question**
  - $A \perp\!\!\!\perp B \mid C$?
  - Are $\mathbf{A}$ and $\mathbf{B}$ d-separated by C?
- **Answer:**
  - An active path exists

    $\Rightarrow$ **no**

Example





- **Question**
  - $A \perp\!\!\!\perp B \mid F$?
  - Are **A** and **B** d-separated by F?

- **Answer:**
  - No active path exists

    $\Rightarrow$ **yes**

Summary

- **Conditional independence (naïve Bayes) assumption**
  - It simplifies computing joint probability

- **Naïve Bayes classifier: simple but often works well in practice**

$$C_k = \underset{C_j}{\operatorname{argmax}} P(C_j) \prod_{i=1}^{n} P(x_i \mid C_j)$$

- **Bayes nets compactly encode joint distributions $P(X_1, X_2, \dots, X_n)$**

$$P(\boldsymbol{X}) = P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{parents}(X_i))$$

- **D-separation determines CI guarantees from graph structure alone**
  - Without doing this mathematically, which is time consuming!

Thank You !